8

Large-Sample Theory

To this point, most of the statistical results in this book concern properties that hold in some exact sense. An estimator is either sufficient or not, unbiased or not, Bayes or not. If exact properties are impractical or not available, statisticians often rely on approximations. This chapter gives several of the most basic results from probability theory used to derive approximations. Several notions of convergence for random variables and vectors are introduced, and various limit theorems are presented. These results are used in this chapter and later to study and compare the performance of various estimators in large samples.

8.1 Convergence in Probability

Our first notion of convergence holds if the variables involved are close to their limit with high probability.

Definition 8.1. *A sequence of random variables Y_n* converges in probability *to a random variable Y as $n \to \infty$, written*

$$Y_n \xrightarrow{p} Y,$$

if for every $\epsilon > 0$,

$$P\big(|Y_n - Y| \geq \epsilon\big) \to 0$$

as $n \to \infty$.

Theorem 8.2 (Chebyshev's Inequality). *For any random variable X and any constant $a > 0$,*

$$P\big(|X| \geq a\big) \leq \frac{EX^2}{a^2}.$$

R.W. Keener, *Theoretical Statistics: Topics for a Core Course*, Springer Texts in Statistics,
DOI 10.1007/978-0-387-93839-4_8,

Proof. Regardless of the value of X,

$$I\{|X| \geq a\} \leq X^2/a^2.$$

The result follows by taking expectations. □

Proposition 8.3. *If $E(Y_n - Y)^2 \to 0$ as $n \to \infty$, then $Y_n \xrightarrow{p} Y$.*

Proof. By Chebyshev's inequality, for any $\epsilon > 0$,

$$P\big(|Y_n - Y| \geq \epsilon\big) \leq \frac{E(Y_n - Y)^2}{\epsilon^2} \to 0. \qquad \square$$

Example 8.4. Suppose $X_1, X_2, \dots$ are i.i.d., with mean μ and variance σ^2, and let $\overline{X}_n = (X_1 + \cdots + X_n)/n$. Then

$$E(\overline{X}_n - \mu)^2 = \mathrm{Var}(\overline{X}_n) = \sigma^2/n \to 0,$$

and so $\overline{X}_n \xrightarrow{p} \mu$ as $n \to \infty$. In fact, $\overline{X}_n \xrightarrow{p} \mu$ even when $\sigma^2 = \infty$, provided $E|X_i| < \infty$. This result is called the *weak law of large numbers.*

Proposition 8.5. *If f is continuous at c and if $Y_n \xrightarrow{p} c$, then $f(Y_n) \xrightarrow{p} f(c)$.*

Proof. Because f is continuous at c, given any $\epsilon > 0$, there exists $\delta_\epsilon > 0$ such that $|f(y) - f(c)| < \epsilon$ whenever $|y - c| < \delta_\epsilon$. Thus

$$P\big(|Y_n - c| < \delta_\epsilon\big) \leq P\big(|f(Y_n) - f(c)| < \epsilon\big),$$

which implies

$$P\big(|f(Y_n) - f(c)| \geq \epsilon\big) \leq P\big(|Y_n - c| \geq \delta_\epsilon\big) \to 0. \qquad \square$$

In statistics there is a family of distributions of interest, indexed by a parameter $\theta \in \Omega$, and the symbol $\xrightarrow{P_\theta}$ is used to denote convergence in probability with P_θ as the underlying probability measure.

Definition 8.6. *A sequence of estimators δ_n, $n \geq 1$, is* consistent *for $g(\theta)$ if for any $\theta \in \Omega$,*

$$\delta_n \xrightarrow{P_\theta} g(\theta)$$

as $n \to \infty$.

If $R(\theta, \delta_n) = E_\theta\big(\delta_n - g(\theta)\big)^2$ is the mean squared error, or risk, of δ_n under squared error loss, then by Proposition 8.3, δ_n will be consistent if $R(\theta, \delta_n) \to 0$ as $n \to \infty$, for any $\theta \in \Omega$. Letting $b_n(\theta) = E_\theta \delta_n - g(\theta)$, called the *bias* of δ_n,

$$R(\theta, \delta_n) = b_n^2(\theta) + \mathrm{Var}_\theta(\delta_n),$$

and so sufficient conditions for consistency are that $b_n(\theta) \to 0$ and $\mathrm{Var}_\theta(\delta_n) \to 0$ as $n \to \infty$, for all $\theta \in \Omega$.

Convergence in probability extends directly to higher dimensions. If Y, Y_1, $Y_2, \dots$ are random vectors in $\mathbb{R}^p$, then Y_n converges in probability to Y, written $Y_n \xrightarrow{p} Y$, if for every $\epsilon > 0$, $P(\|Y_n - Y\| > \epsilon) \to 0$ as $n \to \infty$. Equivalently, $Y_n \xrightarrow{p} Y$ if $[Y_n]_i \xrightarrow{p} [Y]_i$ as $n \to \infty$ for $i = 1, \dots, p$. Proposition 8.5 also holds as stated, with the same proof, for random vectors Y_n and $c \in \mathbb{R}^p$, with f a vector-valued function, $f : \mathbb{R}^p \to \mathbb{R}^q$. For instance, since addition and multiplication are continuous functions from $\mathbb{R}^2 \to \mathbb{R}$, if $X_n \xrightarrow{p} a$ and $Y_n \xrightarrow{p} b$ as $n \to \infty$, then

$$X_n + Y_n \xrightarrow{p} a + b \text{ and } X_n Y_n \xrightarrow{p} ab, \tag{8.1}$$

as $n \to \infty$.

8.2 Convergence in Distribution

If a sequence of estimators δ_n is consistent for $g(\theta)$, then the distribution of the error $\delta_n - g(\theta)$ must concentrate around zero as n increases. But convergence in probability will not tell us how rapidly this concentration occurs or the shape of the error distribution after suitable magnification. For this, the following notion of convergence in distribution is more appropriate.

Definition 8.7. *A sequence of random variables Y_n, $n \geq 1$, with cumulative distribution functions H_n,* converges in distribution *(or law) to a random variable Y with cumulative distribution function H if*

$$H_n(y) \to H(y)$$

as $n \to \infty$ whenever H is continuous at y. For notation we write $Y_n \Rightarrow Y$ or $Y_n \Rightarrow P_Y$.

One aspect of this definition that may seem puzzling at first is that pointwise convergence of the cumulative distribution functions only has to hold at continuity points of H. Here is a simple example that should make this seem more natural.

Example 8.8. Suppose $Y_n = 1/n$, a degenerate random variable, and that Y is always zero. Then

$$H_n(y) = P(Y_n \leq y) = I\{1/n \leq y\}.$$

If $y > 0$, then $H_n(y) = I\{1/n \leq y\} \to 1$ as $n \to \infty$, for eventually $1/n$ will be less than y. If $y \leq 0$, then $H_n(y) = I\{1/n \leq y\} = 0$ for all n, and so $H_n(y) \to 0$ as $n \to \infty$. Because $H(y) = P(Y \leq y) = I\{0 \leq y\}$, comparisons with the limits just obtained show that $H_n(y) \to H(y)$ if $y \neq 0$. But $H_n(0) = 0 \to 0 \neq 1 = H(0)$. In this example, $Y_n \Rightarrow Y$, but the cumulative distribution functions $H_n(y)$ do not converge to $H(y)$ when $y = 0$, a discontinuity point of H.

Theorem 8.9. *Convergence in distribution, $Y_n \Rightarrow Y$, holds if and only if $Ef(Y_n) \to Ef(Y)$ for all bounded continuous functions f.*

Remark 8.10. The convergence of expectations in this theorem is often taken as the definition for convergence in distribution. One advantage of this as a definition is that it generalizes easily to random vectors. Extensions to more abstract objects, such as random functions, are even possible.

Corollary 8.11. *If g is a continuous function and $Y_n \Rightarrow Y$, then*

$$g(Y_n) \Rightarrow g(Y).$$

Proof. If f is bounded and continuous, then $f \circ g$ is also bounded and continuous. Since $Y_n \Rightarrow Y$,

$$Ef\bigl(g(Y_n)\bigr) \to Ef\bigl(g(Y)\bigr).$$

Because f is arbitrary, this shows that the second half of Theorem 8.9 holds for the induced sequence $g(Y_n)$ and $g(Y)$. So by the equivalence, $g(Y_n) \Rightarrow g(Y)$. ☐

For convergence in distribution, the central limit theorem is our most basic tool. For a derivation and proof, see Appendix A.7, or any standard text on probability.

Theorem 8.12 (Central Limit Theorem). *Suppose $X_1, X_2, \dots$ are i.i.d. with common mean μ and variance σ^2. Take $\overline{X}_n = (X_1 + \cdots + X_n)/n$. Then*

$$\sqrt{n}(\overline{X}_n - \mu) \Rightarrow N(0, \sigma^2).$$

As an application of this result, let H_n denote the cumulative distribution function of $\sqrt{n}(\overline{X}_n - \mu)$ and note that

$$\begin{aligned} P(\mu - a/\sqrt{n} < \overline{X}_n \le \mu + a/\sqrt{n}) &= P\bigl(-a < \sqrt{n}(\overline{X}_n - \mu) \le a\bigr) \\ &= H_n(a) - H_n(-a) \\ &\to \Phi(a/\sigma) - \Phi(-a/\sigma). \end{aligned}$$

This information about the distribution of $\overline{X}_n$ from the central limit theorem is more detailed than information from the weak law of large numbers, that $\overline{X}_n \xrightarrow{p} \mu$.

The central limit theorem is certainly one of the most useful and celebrated results in probability and statistics, and it has been extended in numerous ways. Theorems 9.27 and 9.40 provide extensions to averages of i.i.d. random vectors and martingales, respectively. Other extensions concern situations in which the summands are independent but from different distributions or weakly dependent in a suitable sense. In addition, some random variables will be approximately normal because their difference from a variable in one of these central limit theorems converges to zero, an approach used repeatedly later in this book. Results bounding the error in the central limit theorem have also been derived. With the assumptions of Theorem 8.12, the Berry–Esséen theorem, given as Theorem 16.5.1 of Feller (1971), states that

$$\left|P\big(\sqrt{n}(\overline{X}_n - \mu) \le x\big) - \Phi(x/\sigma)\right| \le \frac{3E|X_1 - \mu|^3}{\sigma^3\sqrt{n}}. \tag{8.2}$$

The next result begins to develop a calculus for convergence of random variables combining convergence in distribution with convergence in probability.

Theorem 8.13. *If $Y_n \Rightarrow Y$, $A_n \xrightarrow{p} a$, and $B_n \xrightarrow{p} b$, then*

$$A_n + B_n Y_n \Rightarrow a + bY.$$

The central limit theorem stated only provides direct information about distributions of averages. Many estimators in statistics are not exactly averages, but can be related to averages in some fashion. In some of these cases, clever use of the central limit theorem still provides a limit theorem for an estimator's distribution. A first possibility would be for variables that are smooth functions of an average and can be written as $f(\overline{X}_n)$. The Taylor approximation

$$f(\overline{X}_n) \approx f(\mu) + f'(\mu)(\overline{X}_n - \mu)$$

with the central limit theorem motivates the following proposition.

Proposition 8.14 (Delta Method). *With the assumptions in the central limit theorem, if f is differentiable at μ, then*

$$\sqrt{n}\big(f(\overline{X}_n) - f(\mu)\big) \Rightarrow N\big(0, [f'(\mu)]^2\sigma^2\big).$$

Proof. For convenience, let us assume that f has a continuous derivative[1] and write

$$f(\overline{X}_n) = f(\mu) + f'(\mu_n)(\overline{X}_n - \mu),$$

where μ_n is an intermediate point lying between $\overline{X}_n$ and μ. Since $|\mu_n - \mu| \le |\overline{X}_n - \mu|$ and $\overline{X}_n \xrightarrow{p} \mu$, $\mu_n \xrightarrow{p} \mu$, and since f' is continuous, $f'(\mu_n) \xrightarrow{p} f'(\mu)$ by Proposition 8.5. If $Z \sim N(0, \sigma^2)$, then $\sqrt{n}(\overline{X}_n - \mu) \Rightarrow Z \sim N(0, \sigma^2)$ by the central limit theorem. Thus by Theorem 8.13,

$$\sqrt{n}\big(f(\overline{X}_n) - f(\mu)\big) = f'(\mu_n)\big[\sqrt{n}(\overline{X}_n - \mu)\big] \Rightarrow f'(\mu)Z \sim N\big(0, [f'(\mu)]^2\sigma^2\big).$$

This use of Taylor's theorem to approximate distributions is called the *delta method.* ☐

[1] A proof under the stated condition takes a bit more care; one approach is given in the discussion following Proposition 8.24.

By Theorem 8.9, if $X_n \Rightarrow X$ and f is bounded and continuous, $Ef(X_n) \to Ef(X)$. If f is continuous but unbounded, convergence of $Ef(X_n)$ may fail. The theorem below shows that convergence will hold if the variables are uniformly integrable according to the following definition.

Definition 8.15. *Random variables X_n, $n \ge 1$, are* uniformly integrable *if*

$$\sup_{n\ge 1} E\bigl[|X_n| I\{|X_n| \ge t\}\bigr] \to 0,$$

as $t \to \infty$.

Because $E|X_n| \le t + E\bigl[|X_n| I\bigl\{|X_n| \ge t\bigr\}\bigr]$, if $\sup_{n\ge 1} E\bigl[|X_n| I\{|X_n| \ge t\}\bigr]$ is finite for some t, $\sup_n E|X_n| < \infty$. Thus uniform integrability implies $\sup_n E|X_n| < \infty$. But the converse can fail. If $Y_n \sim \text{Bernoulli}(1/n)$ and $X_n = nY_n$, then $E|X_n| = 1$ for all n, but the variables X_n, $n \ge 1$, are not uniformly integrable.

Theorem 8.16. *If $X_n \Rightarrow X$, then $E|X| \le \liminf E|X_n|$. If X_n, $n \ge 1$, are uniformly integrable and $X_n \Rightarrow X$, then $EX_n \to EX$. If X and X_n, $n \ge 1$, are nonnegative and integrable with $X_n \Rightarrow X$ and $EX_n \to EX$, then X_n, $n \ge 1$, are uniformly integrable.*

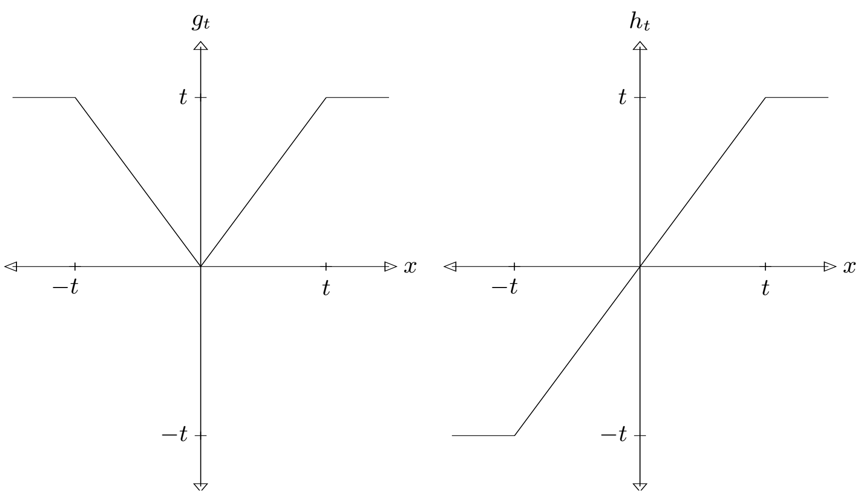

Fig. 8.1. Functions g_t and h_t.

Proof. For $t > 0$, define functions[2] $g_t(x) = |x| \wedge t$ and $h_t(x) = -t \vee (x \wedge t)$, pictured in Figure 8.1. These functions are bounded and continuous, and so if

[2] Here $x \wedge y \overset{\text{def}}{=} \min\{x, y\}$ and $x \vee y \overset{\text{def}}{=} \max\{x, y\}$.

$X_n \Rightarrow X$, $Eg_t(X_n) \to Eg_t(X)$ and $Eh_t(X_n) \to Eh_t(X)$. For the first assertion in the theorem,

$$\liminf E|X_n| \geq \liminf E|X_n| \wedge t = E|X| \wedge t,$$

and the right-hand side increases to $E|X|$ as $t \to \infty$ by monotone convergence (Problem 1.25).

For the second assertion, by uniform integrability and the first result, $E|X| < \infty$. Since $|X_n - h_t(X)| \leq |X_n| I\{|X_n| \geq t\}$,

$$\begin{aligned}\limsup |EX_n - EX| &\leq \limsup |Eh_t(X_n) - Eh_t(X)| + E\,|X - h_t(X)| \\ &\quad + \sup E\,|X_n - h_t(X_n)| \\ &\leq E\,|X - h_t(X)| + \sup E\left[|X_n| I\big\{|X_n| \geq t\big\}\right],\end{aligned}$$

which decreases to zero as $t \to \infty$.

For the final assertion, since the variables are nonnegative with $EX_n \to EX$ and $Eg_t(X_n) \to Eg_t(X)$, then for any $t > 0$,

$$E(X_n - t)^+ = EX_n - Eg_t(X_n) \to EX - Eg_t(X) = E(X - t)^+.$$

Using this, since $xI\{x \geq 2t\} \leq 2(x - \alpha)^+$, $x > 0$,

$$\limsup E\big[X_n I\{X_n \geq 2t\}\big] \leq \limsup 2E(X_n - t)^+ = 2E(X - t)^+.$$

By dominated convergence $E(X - t)^+ \to 0$ as $t \to \infty$. Thus

$$\lim_{t\to\infty} \limsup E\big[|X_n| I\{X_n \geq 2t\}\big].$$

Uniform integrability follows fairly easily from this (see Problem 8.9). ☐

8.3 Maximum Likelihood Estimation

Suppose data vector X has density p_θ. This density, evaluated at X and viewed as a function of θ, $L(\theta) = p_\theta(X)$, is called the *likelihood function*, and the value $\hat{\theta} = \hat{\theta}(X)$ maximizing $L(\cdot)$ is called the *maximum likelihood estimator* of θ. The maximum likelihood estimator of $g(\theta)$ is defined[3] to be $g(\hat{\theta})$. For explicit calculation it is often convenient to maximize the log-likelihood function, $l(\theta) = \log L(\theta)$. instead of $L(\cdot)$.

Example 8.17. Suppose the density for our data X comes from a canonical one-parameter exponential family with density

$$p_\eta(x) = \exp\big\{\eta T(x) - A(\eta)\big\} h(x).$$

[3] It is not hard to check that this definition remains consistent if different parameters are used to specify the model.

Then the maximum likelihood estimator $\hat{\eta}$ of η maximizes

$$l(\eta) = \log p_\eta(X) = \eta T - A(\eta) + \log h(X).$$

Because $l''(\eta) = -A''(\eta) = -\mathrm{Var}_\eta(T) < 0$, the $\hat{\eta}$ is typically[4] the unique solution of

$$0 = l'(\eta) = T - A'(\eta).$$

Letting ψ denote the inverse function of A',

$$\hat{\eta} = \psi(T).$$

If our data are a random sample $X_1, \dots, X_n$ with common density p_η, then the joint density is $\prod_{i=1}^n p_\eta(x_i)$ and the log-likelihood is

$$l(\eta) = \eta \sum_{i=1}^n T(X_i) - nA(\eta) + \log \prod_{i=1}^n h(X_i).$$

The maximum likelihood estimator $\hat{\eta}$ solves

$$0 = l'(\eta) = \sum_{i=1}^n T(X_i) - nA'(\eta),$$

and so

$$\hat{\eta} = \psi(\overline{T}), \quad \text{where } \overline{T} = \frac{1}{n}\sum_{i=1}^n T(X_i).$$

It is interesting to note that the maximum likelihood estimator for the mean of T, $E_\eta T(X_i) = A'(\eta)$, is

$$A'(\hat{\eta}) = A'\bigl(\psi(\overline{T})\bigr) = \overline{T}.$$

The maximum likelihood estimator here is an unbiased function of the complete sufficient statistic; therefore, it is also UMVU. But in general maximum likelihood estimators may have some bias.

Since the maximum likelihood estimator in this example is a function of an average of i.i.d. variables, its asymptotic distribution can be determined using the delta method, Proposition 8.14. By the implicit function theorem, ψ has derivative $(1/A'') \circ \psi$. This derivative evaluated at $A'(\eta) = E_\eta T(X_i)$ is

$$\frac{1}{A''\bigl[\psi\bigl(A'(\eta)\bigr)\bigr]} = \frac{1}{A''(\eta)}.$$

Because $\mathrm{Var}_\eta\bigl(T(X_i)\bigr) = A''(\eta)$, by Proposition 8.14

[4] Examples are possible in which $l(\cdot)$ is strictly increasing or strictly decreasing. The equation here holds whenever $T \in \eta'(\Xi)$.

$$\sqrt{n}(\hat{\eta} - \eta) \Rightarrow N\bigl(0, 1/A''(\eta)\bigr). \tag{8.3}$$

Note that since the Fisher information from each observation is $A''(\eta)$, by the Cramér–Rao lower bound, if an estimator $\tilde{\eta}$ is unbiased for η, then

$$\operatorname{Var}_\eta\bigl[\sqrt{n}(\tilde{\eta} - \eta)\bigr] = n\operatorname{Var}_\eta(\tilde{\eta}) \geq \frac{1}{A''(\eta)}.$$

So (8.3) can be interpreted as showing that $\hat{\eta}$ achieves the Cramér–Rao lower bound in an asymptotic sense. For this reason, $\hat{\eta}$ is considered asymptotically efficient. A rigorous treatment of asymptotic efficiency is delicate and technical; a few of the main developments are given in Section 16.6.

8.4 Medians and Percentiles

Let $X_1, \dots, X_n$ be random variables. These variables, arranged in increasing order, $X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$, are called *order statistics*. The first order statistic $X_{(1)}$ is the smallest value, $X_{(1)} = \min\{X_1, \dots, X_n\}$, and the last order statistic $X_{(n)}$ is the largest value, $X_{(n)} = \max\{X_1, \dots, X_n\}$. The *median* is the middle order statistic when n is odd, or (by convention) the average of the two middle order statistics when n is even:

$$\tilde{X} = \begin{cases} X_{(m)}, & n = 2m - 1; \\ \frac{1}{2}\bigl(X_{(m)} + X_{(m+1)}\bigr), & n = 2m. \end{cases}$$

The median $\tilde{X}$ and mean $\overline{X}$ are commonly used to describe the center or overall location of the variables $X_1, \dots, X_n$. One possible advantage for the median is that it will not be influenced by a few extreme values. For instance, if the data are $(1,2,3,4,5)$, then both $\tilde{X}$ and $\overline{X}$ are 3. But if the data are $(1,2,3,4,500)$, $\tilde{X}$ is still 3, but $\overline{X} = 102$. If we view them as estimators, it is also natural to want to compare the error distributions of $\overline{X}$ and $\tilde{X}$. For a random sample, the error distribution of $\overline{X}$ can be approximated using the central limit theorem. In what follows, we derive an analogous result for $\tilde{X}$.

Assume now that $X_1, X_2, \dots$ are i.i.d. with common cumulative distribution function F, and let $\tilde{X}_n$ be the median of the first n observations. For regularity, assume that F has a unique median θ, so $F(\theta) = 1/2$, and that $F'(\theta)$ exists and is finite and positive. Let us try to approximate

$$P\bigl(\sqrt{n}(\tilde{X}_n - \theta) \leq a\bigr) = P(\tilde{X}_n \leq \theta + a/\sqrt{n}).$$

Define

$$S_n = \#\{i \leq n : X_i \leq \theta + a/\sqrt{n}\}.$$

The key to this derivation is the observation that $\tilde{X}_n \leq \theta + a/\sqrt{n}$ if and only if $S_n \geq m$. Also, by viewing observation i as a success if $X_i \leq \theta + a/\sqrt{n}$, it is evident that

$$S_n \sim \text{Binomial}\big(n, F(\theta + a/\sqrt{n})\big).$$

The next step involves normal approximation for the distribution of S_n. First note that if $Y_n \sim \text{Binomial}(n, p)$, then Y_n/n can be viewed as the average of n i.i.d. Bernoulli variables. Therefore by the central limit theorem,

$$\sqrt{n}\left(\frac{Y_n}{n} - p\right) = \frac{Y_n - np}{\sqrt{n}} \Rightarrow N\left(0, p(1-p)\right),$$

and hence

$$\begin{aligned} P\left(\frac{Y_n - np}{\sqrt{n}} > y\right) &= 1 - P\left(\frac{Y_n - np}{\sqrt{n}} \le y\right) \\ &\to 1 - \Phi\left(\frac{y}{\sqrt{p(1-p)}}\right) = \Phi\left(\frac{-y}{\sqrt{p(1-p)}}\right), \end{aligned}$$

as $n \to \infty$. In fact, this approximation for the binomial distribution holds uniformly in y and uniformly for p in any compact subset of $(0,1)$.[5]

The normal approximation for the binomial distribution just discussed gives

$$\begin{aligned} &P\big(\sqrt{n}(\tilde{X}_n - \theta) \le a\big) \\ &\quad = P(S_n > m-1) \\ &\quad = P\left(\frac{S_n - nF(\theta + a/\sqrt{n})}{\sqrt{n}} > \frac{m - 1 - nF(\theta + a/\sqrt{n})}{\sqrt{n}}\right) \\ &\quad = \Phi\left(\frac{\big[nF(\theta + a/\sqrt{n}) - m + 1\big]/\sqrt{n}}{\sqrt{F(\theta + a/\sqrt{n})\big(1 - F(\theta + a/\sqrt{n})\big)}}\right) + o(1). \end{aligned} \tag{8.4}$$

Here "$o(1)$" is used to denote a sequence that tends to zero as $n \to \infty$. See Section 8.6 for a discussion of notation and various notions of scales of magnitude. Since F is continuous at θ,

$$\sqrt{F(\theta + a/\sqrt{n})\big(1 - F(\theta + a/\sqrt{n})\big)} \to 1/2,$$

as $n \to \infty$. And because F is differentiable at θ,

$$\begin{aligned} \frac{nF(\theta + a/\sqrt{n}) - m + 1)}{\sqrt{n}} &= a\frac{F(\theta + a/\sqrt{n}) - F(\theta)}{a/\sqrt{n}} + \frac{nF(\theta) - m + 1}{\sqrt{n}} \\ &= a\frac{F(\theta + a/\sqrt{n}) - F(\theta)}{a/\sqrt{n}} + \frac{1}{2\sqrt{n}} \to aF'(\theta). \end{aligned}$$

[5] "Uniformity" here means that the difference between the two sides will tend to zero as $n \to \infty$, even if y and p both vary with n, provided p stays away from zero and one ($\limsup p < 1$ and $\liminf p > 0$). This can be easily proved using the Berry–Esséen bound (8.2).

Since the numerator and denominator of the argument of Φ in (8.4) both converge,

$$P\big(\sqrt{n}(\tilde{X}_n - \theta) \le a\big) \to \Phi\big(2aF'(\theta)\big).$$

The limit here is the cumulative distribution function for the normal distribution with mean zero and variance $1/\big(4[F'(\theta)]^2\big)$ evaluated at a, and so

$$\sqrt{n}(\tilde{X}_n - \theta) \Rightarrow N\left(0, \frac{1}{4[F'(\theta)]^2}\right). \tag{8.5}$$

A similar derivation leads to the following central limit theorem for other quantiles.

Theorem 8.18. *Let $X_1, X_2, \dots$ be i.i.d. with common cumulative distribution function F, let $\gamma \in (0,1)$, and let $\tilde{\theta}_n$ be the $\lfloor \gamma n \rfloor$th order statistic for $X_1, \dots, X_n$ (or a weighted average of the $\lfloor \gamma n \rfloor$th and $\lceil \gamma n \rceil$th order statistics).*[6] *If $F(\theta) = \gamma$, and if $F'(\theta)$ exists and is finite and positive, then*

$$\sqrt{n}(\tilde{\theta}_n - \theta) \Rightarrow N\left(0, \frac{\gamma(1-\gamma)}{\big[F'(\theta)\big]^2}\right),$$

as $n \to \infty$.

8.5 Asymptotic Relative Efficiency

A comparison of the mean and median will only be natural if they both estimate the same parameter. In a location family this will happen naturally if the error distribution is symmetric. So let us assume that our data are i.i.d. and have common density $f(x - \theta)$ with f symmetric about zero, $f(u) = f(-u)$, $u \in \mathbb{R}$. Then $P_\theta(X_i < \theta) = P_\theta(X_i > \theta) = 1/2$, and $E_\theta X_i = \theta$ (provided the mean exists). By the central limit theorem,

$$\sqrt{n}(\overline{X}_n - \theta) \Rightarrow N(0, \sigma^2),$$

where

$$\sigma^2 = \int x^2 f(x)\, dx,$$

and by (8.5),

$$\sqrt{n}(\tilde{X}_n - \theta) \Rightarrow N\left(0, \frac{1}{4f^2(0)}\right).$$

(Here we naturally take $f(0) = F'(0)$.) Suppose f is the standard normal density, $f(x) = e^{-x^2/2}/\sqrt{2\pi}$. Then $\sigma^2 = 1$ and $1/\big(4f^2(0)\big) = \pi/2$. Since the

[6] Here $\lfloor x \rfloor$, called the *floor* of x, is the largest integer y with $y \le x$. Also, $\lceil x \rceil$ is the smallest integer $y \ge x$, called the *ceiling* of x.

variance of the limiting distribution is larger for the median than the mean, the median is less efficient than the mean. To understand the import of this difference in efficiency, define $m = m_n = \lfloor \pi n/2 \rfloor$, and note that $\sqrt{n/m} \to \sqrt{2/\pi}$ as $n \to \infty$. Using Theorem 8.13,

$$\sqrt{n}(\tilde{X}_m - \theta) = \sqrt{\frac{n}{m}}\sqrt{m}(\tilde{X}_m - \theta) \Rightarrow N(0,1).$$

This shows that the error distribution for the median of m observations is approximately the same as the error distribution for the mean of n observations. As $n \to \infty$, $m/n \to \pi/2$, and this limiting ratio $\pi/2$ is called the *asymptotic relative efficiency* (ARE) of the mean $\overline{X}_n$ with respect to the median $\tilde{X}_n$. In general, if $\hat{\theta}_n$ and $\tilde{\theta}_n$ are sequences of estimators, and if

$$\sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow N(0, \sigma^2_{\hat{\theta}})$$

and

$$\sqrt{n}(\tilde{\theta}_n - \theta) \Rightarrow N(0, \sigma^2_{\tilde{\theta}}),$$

then the asymptotic relative efficiency of $\hat{\theta}_n$ with respect to $\tilde{\theta}_n$ is $\sigma^2_{\tilde{\theta}}/\sigma^2_{\hat{\theta}}$. This relative efficiency can be interpreted as the ratio of sample sizes necessary for comparable error distributions.

In our first comparison of the mean and median the data were a random sample from $N(\theta, 1)$. In this case, the mean is UMVU, so it should be of no surprise that it is more efficient than the median. If instead

$$f(x) = \frac{1}{2}e^{-|x|},$$

then

$$\sigma^2 = \int x^2 \frac{1}{2} e^{-|x|}\,dx = \int_0^\infty x^2 e^{-x}\,dx = \Gamma(3) = 2! = 2.$$

So here $\sqrt{n}(\overline{X}_n - \theta) \Rightarrow N(0,2)$, $\sqrt{n}(\tilde{X}_n - \theta) \Rightarrow N(0,1)$, and the asymptotic relative efficiency of $\overline{X}_n$ with respect to $\tilde{X}_n$ is $1/2$. Now the median is more efficient than the mean, and roughly twice as many observations will be needed for a comparable error distribution if the mean is used instead of the median. In this case, the median is the maximum likelihood estimator of θ. Later results in Sections 9.3 and 16.6 show that maximum likelihood estimators are generally fully efficient.

Example 8.19. Suppose $X_1, \dots, X_n$ is a random sample from $N(\theta, 1)$, and we are interested in estimating

$$p = P_\theta(X_i \le a) = \Phi(a - \theta).$$

One natural estimator is

$$\hat{p} = \Phi(a - \overline{X}),$$

where $\overline{X} = (X_1 + \cdots + X_n)/n$. (This is the maximum likelihood estimator.) Another natural estimator is the proportion of the observations that are at most a,

$$\tilde{p} = \frac{1}{n}\#\{i \le n : X_i \le a\} = \frac{1}{n}\sum_{i=1}^{n} I\{X_i \le a\}.$$

By the central limit theorem,

$$\sqrt{n}(\tilde{p} - p) \Rightarrow N(0, \tilde{\sigma}^2),$$

as $n \to \infty$, where

$$\tilde{\sigma}^2 = \operatorname{Var}_\theta\bigl(I\{X_i \le a\}\bigr) = \Phi(a-\theta)\bigl(1 - \Phi(a-\theta)\bigr).$$

Because the first estimator is a function of the average $\overline{X}$, by the delta method, Proposition 8.14,

$$\sqrt{n}(\hat{p} - p) \Rightarrow N(0, \hat{\sigma}^2),$$

as $n \to \infty$, where

$$\hat{\sigma}^2 = \left[\frac{d}{dx}\Phi(a-x)\Big|_{x=\theta}\right]^2 = \phi^2(a-\theta).$$

The asymptotic relative efficiency of $\hat{p}$ with respect to $\tilde{p}$ is

$$\text{ARE} = \frac{\Phi(a-\theta)\bigl(1-\Phi(a-\theta)\bigr)}{\phi^2(a-\theta)}.$$

In this example, the asymptotic relative efficiency depends on the unknown parameter θ. When $\theta = a$, ARE $= \pi/2$, and the ARE increases without bound as $|\theta - a|$ increases. Note, however, that $\tilde{p}$ is a sensible estimator even if the stated model is wrong, provided the data are indeed i.i.d. In contrast, $\hat{p}$ is only reasonable if the model is correct. Gains in efficiency using $\hat{p}$ should be balanced against the robustness of $\tilde{p}$ to departures from the model.

8.6 Scales of Magnitude

In many asymptotic calculations it is convenient to have a standard notation indicating orders of magnitudes of variables in limiting situations. We begin with a definition for sequences of constants.

Definition 8.20. *Let a_n and b_n, $n \ge 1$, be constants. Then*

1. *$a_n = o(b_n)$ as $n \to \infty$ means that $a_n/b_n \to 0$ as $n \to \infty$;*
2. *$a_n = O(b_n)$ as $n \to \infty$ means that $|a_n/b_n|$ remains bounded, i.e., that $\limsup_{n\to\infty} |a_n/b_n| < \infty$; and*

3. $a_n \sim b_n$ *means that* $a_n/b_n \to 1$ *as* $n \to \infty$.

Thus $a_n = o(b_n)$ when a_n is of smaller order of magnitude than b_n, $a_n = O(b_n)$ when the magnitude of a_n is at most comparable to the magnitude of b_n, and $a_n \sim b_n$ when a_n is asymptotic to b_n. Note that $a_n = o(1)$ means that $a_n \to 0$.

Large oh and small oh notation may also be used in equations or inequalities. For instance, $a_n = b_n + O(c_n)$ means that $a_n - b_n = O(c_n)$, and $a_n \le b_n + o(c_n)$ means that $a_n \le b_n + d_n$ for some sequence d_n with $d_n = o(c_n)$. Exploiting this idea, $a_n \sim b_n$ can be written as $a_n = b_n\bigl(1 + o(1)\bigr)$.

Although Definition 8.20 is stated for sequences indexed by a discrete variable n, analogous notation can be used for functions indexed by a continuous variable x. For instance, $a(x) = o\bigl(b(x)\bigr)$ as $x \to x_0$ would mean that $a(x)/b(x) \to 0$ as $x \to x_0$. The limit x_0 here could be finite or infinite. As an example, if f has two derivatives at x, then the two-term Taylor expansion for f can be expressed as

$$f(x+\epsilon) = f(x) + \epsilon f'(x) + \frac{1}{2}\epsilon^2 f''(x) + o(\epsilon^2)$$

as $\epsilon \to 0$. If f''' is exists and is finite at x, this can be strengthened to

$$f(x+\epsilon) = f(x) + \epsilon f'(x) + \frac{1}{2}\epsilon^2 f''(x) + O(\epsilon^3)$$

as $\epsilon \to 0$.

In the following stochastic extension, the basic idea is that the original notion can fail, but only on a set with arbitrarily small probability.

Definition 8.21. *Let* X_n *and* Y_n, $n \ge 1$, *be random variables, and let* b_n, $n \ge 1$, *be constants. Then*

1. $X_n = o_p(b_n)$ *as* $n \to \infty$ *means that* $X_n/b_n \xrightarrow{p} 0$ *as* $n \to \infty$;
2. $X_n = O_p(1)$ *as* $n \to \infty$ *means that*

$$\sup_n P\bigl(|X_n| > K\bigr) \to 0$$

as $K \to \infty$; *and*

3. $X_n = O_p(b_n)$ *means that* $X_n/b_n = O_p(1)$ *as* $n \to \infty$.

The definition for $O_p(1)$ is equivalent to a notion called *tightness* for the distributions of the X_n. Tightness is necessary for convergence in distribution, and so, if $X_n \Rightarrow X$, then $X_n = O_p(1)$.

Here are a few useful propositions about stochastic scales of magnitude.

Proposition 8.22. *If* $X_n = O_p(a_n)$ *and* $Y_n = O_p(b_n)$, *then*

$$X_n Y_n = O_p(a_n b_n).$$

Also, if $\alpha > 0$ and $X_n = O_p(a_n)$, then $X_n^\alpha = O_p(a_n^\alpha)$. Similarly, if $X_n = O_p(a_n)$, $\alpha > 0$, and $Y_n = o_p(b_n)$, then

$$X_n Y_n = o_p(a_n b_n) \text{ and } Y_n^\alpha = o_p(b_n^\alpha).$$

Proposition 8.23. *Let α and β be constants with $\alpha > 0$. If $E|X_n|^\alpha = O(n^\beta)$ as $n \to \infty$, then $X_n = O_p(n^{\beta/\alpha})$ as $n \to \infty$.*

Proposition 8.24. *If $X_n = O_p(a_n)$ with $a_n \to 0$, and if $f(\epsilon) = o(\epsilon^\alpha)$ as $\epsilon \to 0$ with $\alpha > 0$, then*

$$f(X_n) = o_p(a_n^\alpha).$$

This result is convenient for delta method derivations such as Proposition 8.14. By the central limit theorem, $\overline{X}_n = \mu + O_p\bigl(1/\sqrt{n}\bigr)$, and by Taylor expansion

$$f(\mu + \epsilon) = f(\mu) + \epsilon f'(\mu) + o(\epsilon)$$

as $\epsilon \to 0$, whenever f is differentiable at μ. So by Proposition 8.24,

$$f\bigl(\overline{X}_n\bigr) - f(\mu) = \bigl(\overline{X}_n - \mu\bigr) f'(\mu) + o_p\bigl(1/\sqrt{n}\bigr),$$

and rearranging terms,

$$\sqrt{n}\bigl(f(\overline{X}_n) - f(\mu)\bigr) = \sqrt{n}\bigl(\overline{X}_n - \mu\bigr) f'(\mu) + o_p(1) \Rightarrow N\bigl(0, [f'(\mu)]^2\sigma^2\bigr).$$

8.7 Almost Sure Convergence[7]

In this section, we consider a notion of convergence for random variables called *almost sure convergence* or *convergence with probability one.*

Definition 8.25. *Random variables $Y_1, Y_2, \ldots$ defined on a common probability space* converge almost surely *to a random variable Y on the same space if*

$$P(Y_n \to Y) = 1.$$

The statistical implications of this mode of convergence are generally similar to the implications of convergence in probability, and in the rest of this book we refer to almost sure convergence only when the distinction seems statistically relevant. To understand the difference between these modes of convergence, introduce

$$M_n = \sup_{k \ge n} |Y_k - Y|,$$

and note that $Y_n \to Y$ if and only if $M_n \to 0$. Now $M_n \to 0$ if for every $\epsilon > 0$, $M_n < \epsilon$ for all n sufficiently large. Define B_ϵ as the event that $M_n < \epsilon$ for all

[7] Results in this section are used only in Chapter 20.

n sufficiently large. An outcome is in B_ϵ if and only if it is in one of the sets $\{M_n < \epsilon\}$, and thus

$$B_\epsilon = \bigcup \{M_n < \epsilon\}.$$

If an outcome gives a convergent sequence, it must be in B_ϵ for every ϵ, and so

$$\{Y_n \to Y\} = \bigcap_{\epsilon>0} B_\epsilon,$$

and we have almost sure convergence if and only if

$$P\left[\bigcap_{\epsilon>0} B_\epsilon\right] = 1.$$

Since the B_ϵ decrease as $\epsilon \to 0$, using the continuity property of probability measures (1.1), this will happen if and only if $P(B_\epsilon) = 1$ for all $\epsilon > 0$. But because the events $\{M_n < \epsilon\}$ increase with n, $P(B_\epsilon) = \lim_{n\to\infty} P(M_n < \epsilon)$. Putting this all together, $Y_n \to Y$ almost surely if and only if for every $\epsilon > 0$, $P(M_n \geq \epsilon) \to 0$, that is, if and only if $M_n \xrightarrow{p} 0$. In words, almost sure convergence means the largest difference after stage n tends to zero in probability as $n \to \infty$.

Example 8.26. If $Y_n \sim \text{Bernoulli}(p_n)$, then $Y_n \xrightarrow{p} 0$ if and only if $p_n \to 0$. Almost sure convergence will also depend on the joint distribution of these variables. If they are independent, then $M_n = \sup_{k\geq n} |Y_n - 0| \sim \text{Bernoulli}(\pi_n)$ with

$$1 - \pi_n = P(M_n = 0) = P(Y_k = 0, k \geq n) = \prod_{k=n}^{\infty} (1 - p_k).$$

This product tends to 1 as $n \to \infty$ if and only if $\sum p_n < \infty$. So in this independent case, $Y_n \to 0$ almost surely if and only if $\sum p_n < \infty$. If instead U is uniformly distributed on $(0, 1)$ and $Y_n = I\{U \leq p_n\}$, then $Y_n \to 0$ almost surely if and only if $p_n \to 0$, that is, if and only if $Y_n \xrightarrow{p} 0$.

The following result is the most famous result on almost sure convergence. For a proof, see Billingsley (1995) or any standard text on probability.

Theorem 8.27 (Strong Law of Large Numbers). *If $X_1, X_2, \ldots$ are i.i.d. with finite mean $\mu = EX_i$, and if $\overline{X}_n = (X_1 + \cdots + X_n)/n$, then $\overline{X}_n \to \mu$ almost surely as $n \to \infty$.*

8.8 Problems[8]

*1. Random variables $X_1, X_2, \ldots$ are called "m-dependent" if X_i and X_j are independent whenever $|i - j| \geq m$. Suppose $X_1, X_2, \ldots$ are m-dependent

[8] Solutions to the starred problems are given at the back of the book.

with $EX_1 = EX_2 = \cdots = \xi$ and $\mathrm{Var}(X_1) = \mathrm{Var}(X_2) = \cdots = \sigma^2 < \infty$. Let $\overline{X}_n = (X_1 + \cdots + X_n)/n$. Show that $\overline{X}_n \xrightarrow{p} \xi$ as $n \to \infty$. Hint: You should be able to bound $\mathrm{Cov}(X_i, X_j)$ and $\mathrm{Var}(\overline{X}_n)$.

*2. Let $X_1, \dots, X_n$ be i.i.d. from an exponential distribution with failure rate λ, and let $M_n = \max\{X_1, \dots, X_n\}$. Is $\log(n)/M_n$ a consistent estimator of λ?

*3. If $X_1, \dots, X_n$ are i.i.d. from the uniform distribution on $(0, \theta)$ with maximum $M_n = \max\{X_1, \dots, X_n\}$, then the UMVU estimator of θ is $\hat{\theta}_n = (n+1)M_n/n$. Determine the limiting distribution of $n(\hat{\theta}_n - \theta)$ as $n \to \infty$.

*4. Let $X_1, \dots, X_n$ be i.i.d. Bernoulli variables with success probability p. Let $\hat{p}_n = (X_1 + \cdots + X_n)/n$.
   a) Show that $\sqrt{n}(\hat{p}_n^2 - p^2) \Rightarrow N\big(0, 4p^3(1-p)\big)$.
   b) Find the UMVU estimator δ_n of $\sigma^2 = 4p^3(1-p)$, the asymptotic variance in (a).
   c) Determine the limiting distribution of $n(\delta_n - \sigma^2)$ when $p = 3/4$. Hint: The maximum likelihood estimator of σ^2 is $\hat{\sigma}^2 = 4\hat{p}_n^3(1-\hat{p}_n)$. Show that $n(\delta_n - \hat{\sigma}_n^2)$ converges in probability to a constant, and use a two-term Taylor expansion to find the limiting distribution of $n(\hat{\sigma}^2 - \sigma^2)$.

*5. Let $X_1, \dots, X_n$ be i.i.d. with common density $f_\theta(x) = (x-\theta)^+ e^{\theta - x}$. Show that $M_n = \min\{X_1, \dots, X_n\}$ is a consistent estimator of θ, and determine the limiting distribution for $\sqrt{n}(M_n - \theta)$.

*6. Prove that if $A_n \xrightarrow{p} 1$ and $Y_n \Rightarrow Y$, then $A_n Y_n \Rightarrow Y$. (This is a special case of Theorem 8.13.)

7. Suppose $X_1, X_2, \dots$ are i.i.d. with common density

$$f(x) = \begin{cases} \dfrac{1}{(1+x)^2}, & x > 0; \\ 0, & \text{otherwise}, \end{cases}$$

and let $M_n = \max\{X_1, \dots, X_n\}$. Show that M_n/n converges in distribution, and give a formula for the limiting distribution function.

8. If $\epsilon > 0$ and $\sup E|X_n|^{1+\epsilon} < \infty$, show that X_n, $n \ge 1$, are uniformly integrable.

9. Suppose $X_1, X_2, \dots$ are integrable and

$$\lim_{t \to \infty} \limsup_{n \to \infty} E\big[|X_n| I\{|X_n| \ge t\}\big] = 0.$$

Show that X_n, $n \ge 1$, are uniformly integrable.

10. Suppose $X_n \Rightarrow X$, $x_n \to x$, and the cumulative distribution function for X is continuous at x. Show that $P(X_n \le x_n) \to P(X \le x)$.

11. Let $X_1, X_2, \dots$ be i.i.d. variables uniformly distributed on $(0, 1)$, and let $\tilde{X}_n$ denote the geometric average of the first n of these variables; that is,

$$\tilde{X}_n = (X_1 \times \cdots \times X_n)^{1/n}.$$

a) Show that $\tilde{X}_n \xrightarrow{p} 1/e$ as $n \to \infty$.
b) Show that $\sqrt{n}(\tilde{X}_n - 1/e)$ converges in distribution, and identify the limit.

12. Let $X_1, X_2, \dots$ be i.i.d. from the uniform distribution on $(1, 2)$, and let H_n denote the harmonic average of the first n variables:

$$H_n = \frac{n}{X_1^{-1} + \cdots + X_n^{-1}}.$$

a) Show that $H_n \xrightarrow{p} c$ as $n \to \infty$, identifying the constant c.
b) Show that $\sqrt{n}(H_n - c)$ converges in distribution, and identify the limit.

13. Show that if $Y_n \xrightarrow{p} c$ as $n \to \infty$, then $Y_n \Rightarrow Y$ as $n \to \infty$. Give the distribution or cumulative distribution function for Y.
14. Let $X_1, X_2, \dots$ be i.i.d. from a uniform distribution on $(0, e)$, and define

$$Y_n = \sqrt[n]{\textstyle\prod_{i=1}^{n^2} X_i}.$$

Show that $Y_n \Rightarrow Y$ as $n \to \infty$, giving the cumulative distribution function for Y.

15. Let $X_1, X_2, \dots$ be i.i.d. from $N(\mu, \sigma^2)$, let $w_1, w_2, \dots$ be positive weights, and define weighted averages

$$Y_n = \frac{\sum_{i=1}^n w_i X_i}{\sum_{i=1}^n w_i}, \qquad n = 1, 2, \dots.$$

a) Suppose $w_k = 1/k$, $k = 1, 2, \dots$. Show that $Y_n \xrightarrow{p} c$, identifying the limiting value c.
b) Suppose $w_k = 1/(2k-1)^2$. Show that $Y_n \Rightarrow Y$, giving the distribution for Y. Hint:

$$\sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} = \frac{\pi^2}{8} \text{ and } \sum_{k=1}^{\infty} \frac{1}{(2k-1)^4} = \frac{\pi^4}{96}.$$

*16. Let $Y_1, \dots, Y_n$ be independent with $Y_i \sim N(\alpha + \beta x_i, \sigma^2)$, $i = 1, \dots, n$, where $x_1, \dots, x_n$ are known constants, and α, β, and σ^2 are unknown parameters. Find the maximum likelihood estimators of these parameters, α, β, and σ^2.

*17. Let $X_1, \dots, X_n$ be jointly distributed. The first variable $X_1 \sim N(0, 1)$, and, for $j = 1, \dots, n-1$, the conditional distribution of X_{j+1} given $X_1 = x_1, \dots, X_j = x_j$ is $N(\rho x_j, 1)$. Find the maximum likelihood estimator of ρ.

18. Distribution theory for order statistics in the tail of the distribution can behave differently than order statistics such as the median, that are near the middle of the distribution. Let $X_1, \dots, X_n$ be i.i.d. from an exponential distribution with unit failure rate.

a) Suppose we are interested in the limiting distribution for $X_{(2)}$, the second order statistic. Naturally, $X_{(2)} \xrightarrow{p} 0$ as $n \to \infty$. For an interesting limit theory we should scale $X_{(2)}$ by an appropriate power of n, but the correct power is not $1/2$. Suppose $x > 0$. Find a value p so that $P(n^p X_{(2)} \le x)$ converges to a value between 0 and 1. (If p is too small, the probability will tend to 1, and if p is too large the probability will tend to 0.)

b) Determine the limiting distribution for $X_{(n)} - \log n$.

*19. Let $X_1, \dots, X_n$ be i.i.d. from an exponential distribution with failure rate θ. Let $\hat{p}_n = \#\{i \le n : X_i \ge 1\}/n$ and $\overline{X}_n = (X_1 + \cdots + X_n)/n$. Determine the asymptotic relative efficiency of $-\log \hat{p}_n$ with respect to $1/\overline{X}_n$.

*20. Let $X_1, \dots, X_n$ be i.i.d. from $N(\theta, \theta)$, with $\theta > 0$ an unknown parameter, and consider estimating $\theta(\theta + 1)$. Determine the asymptotic relative efficiency of $\overline{X}_n(\overline{X}_n + 1)$ with respect to $\delta_n = (X_1^2 + \cdots + X_n^2)/n$, where, as usual, $\overline{X}_n = (X_1 + \cdots + X_n)/n$.

*21. Let Q_n denote the upper quartile (or 75th percentile) for a random sample $X_1, \dots, X_n$ from $N(0, \sigma^2)$. If $\Phi(c) = 3/4$, then $Q_n \xrightarrow{p} c\sigma$, and so $\tilde{\sigma}_n = Q_n/c$ is a consistent estimator of σ. Let $\hat{\sigma}$ be the maximum likelihood estimator of σ. Determine the asymptotic relative efficiency of $\tilde{\sigma}$ with respect to $\hat{\sigma}$.

22. If $X_1, \dots, X_n$ are i.i.d. from $N(\theta, \theta)$, then two natural estimators of θ are the sample mean $\overline{X}$ and the sample variance S^2. Determine the asymptotic relative efficiency of S^2 with respect to $\overline{X}$.

23. Suppose $X_1, \dots, X_n$ are i.i.d. Poisson variables with mean λ and we are interested in estimating $p = P_\lambda(X_i = 0) = e^{-\lambda}$.

a) One estimator for p is the proportion of zeros in the sample, $\tilde{p} = \#\{i \le n : X_i = 0\}/n$. Find the limiting distribution for $\sqrt{n}(\tilde{p} - p)$.

b) Another estimator would be the maximum likelihood estimator $\hat{p}$. Give a formula for $\hat{p}$ and determine the limiting distribution for $\sqrt{n}(\hat{p} - p)$.

c) Find the asymptotic relative efficiency of $\tilde{p}$ with respect to $\hat{p}$.

*24. Suppose $X_1, \dots, X_n$ are i.i.d. $N(0, \sigma^2)$, and let M be the median of $|X_1|, \dots, |X_n|$.

a) Find $c \in \mathbb{R}$ so that $\tilde{\sigma} = cM$ is a consistent estimator of σ.

b) Determine the limiting distribution for $\sqrt{n}(\tilde{\sigma} - \sigma)$.

c) Find the maximum likelihood estimator $\hat{\sigma}$ of σ and determine the limiting distribution for $\sqrt{n}(\hat{\sigma} - \sigma)$.

d) Determine the asymptotic relative efficiency of $\tilde{\sigma}$ with respect to $\hat{\sigma}$.

25. Suppose $X_1, X_2, \dots$ are i.i.d. from the beta distribution with parameters $\alpha > 0$ and $\beta > 0$. The mean of this distribution is $\mu = \alpha/(\alpha + \beta)$. Solving, $\alpha = \beta\mu/(1 - \mu)$. If β is known, this suggests

$$\tilde{\alpha} = \frac{\beta \overline{X}}{1 - \overline{X}}$$

as a natural estimator for α. Determine the asymptotic relative efficiency of this estimator $\tilde{\alpha}$ with respect to the maximum likelihood estimator $\hat{\alpha}$.

26. Let $X_1, \dots, X_n$ be i.i.d. Poisson with mean λ, and consider estimating

$$g(\lambda) = P_\lambda(X_i = 1) = \lambda e^{-\lambda}.$$

One natural estimator might be the proportion of ones in the sample:

$$\hat{p}_n = \frac{1}{n}\#\{i \le n : X_i = 1\}.$$

Another choice would be the maximum likelihood estimator, $g(\overline{X}_n)$, with $\overline{X}_n$ the sample average.

 a) Find the asymptotic relative efficiency of $\hat{p}_n$ with respect to $g(\overline{X}_n)$.
 b) Determine the limiting distribution of

$$n\big[g(\overline{X}_n) - 1/e\big]$$

 when $\lambda = 1$.

27. Let $X_1, \dots, X_n$ be i.i.d. from $N(\theta, 1)$, and let $U_1, \dots, U_n$ be i.i.d. from a uniform distribution on $(0, 1)$, with all $2n$ variables independent. Define $Y_i = X_iU_i$, $i = 1, \dots, n$. If the X_i and U_i are both observed, then $\overline{X}$ would be a natural estimator for θ. If only the products $Y_1, \dots, Y_n$ are observed, then $2\overline{Y}$ may be a reasonable estimator. Determine the asymptotic relative efficiency of $2\overline{Y}$ with respect to $\overline{X}$.
28. Definition 8.21 for $O_p(1)$ does not refer explicitly to limiting values as $n \to \infty$. But in fact the conclusion only depends on the behavior of the sequence for large n. Show that if

$$\limsup_{n\to\infty} P\big(|X_n| > K\big) \to 0$$

as $K \to \infty$, then $X_n = O_p(1)$, so that "sup" in the definition could be changed to "lim sup."

29. Prove Proposition 8.22.
30. *Markov's inequality.* Show that for any constant $c > 0$ and any random variable X,

$$P(|X| \ge c) \le E|X|/c.$$

31. Use Markov's inequality from the previous problem to prove Proposition 8.23.
32. If $X_n \Rightarrow X$ as $n \to \infty$, show that $X_n = O_p(1)$ as $n \to \infty$. Also, show that the converse fails, finding a sequence of random variables X_n that are $O_p(1)$ but do not converge in distribution.
33. Show that if $X_n = O_p(1)$ as $n \to \infty$ and f is a continuous function on $\mathbb{R}$, then $f(X_n) = O_p(1)$ as $n \to \infty$. Also, give an example showing that this result can fail if f is discontinuous at some point x.
34. Let M_n, $n \ge 1$, be positive, integer-valued random variables.
 a) Show that if $M_n \to \infty$ almost surely as $n \to \infty$, and $X_n \to 0$ almost surely as $n \to \infty$, then $X_{M_n} \to 0$ almost surely as $n \to \infty$.

b) Show that if $M_n \xrightarrow{p} \infty$, and $X_n \to 0$ almost surely, then $X_{M_n} \xrightarrow{p} 0$.

35. Let $X_1, X_2, \ldots$ be independent Bernoulli variables with $P(X_n = 1) = 1/n$. Then $X_n \xrightarrow{p} 0$, but almost sure convergence fails. Find positive, integer-valued random variables M_n, $n \geq 1$, such that $M_n \to \infty$ almost surely with $X_{m_n} = 1$. This shows that the almost sure convergence for X_n in the previous problem is essential.